

Prime factorization and numerical fractions

While TSM 099 studies Topic 18. Review: Topic 2, Topic 4.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: prime factorization

1. Write as a product of prime factors.

$$36 = 2^2 \cdot 3^2$$

2. Write as a product of prime factors.

$$48 = 2^4 \cdot 3$$

3. Write as a product of prime factors.

$$60 = 2^2 \cdot 3 \cdot 5$$

4. Write as a product of prime factors.

$$72 = 2^3 \cdot 3^2$$

5. Write as a product of prime factors.

$$84 = 2^2 \cdot 3 \cdot 7$$

6. Write as a product of prime factors.

$$90 = 2 \cdot 3^2 \cdot 5$$

7. Write as a product of prime factors.

$$120 = 2^3 \cdot 3 \cdot 5$$

8. Write as a product of prime factors.

$$180 = 2^2 \cdot 3^2 \cdot 5$$

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Core: simplify using common prime factors

9. Write in lowest terms.

$$\frac{36}{84} = \frac{3}{7}$$

10. Write in lowest terms.

$$\frac{48}{180} = \frac{4}{15}$$

11. Write in lowest terms.

$$\frac{72}{108} = \frac{2}{3}$$

12. Write in lowest terms.

$$\frac{90}{150} = \frac{3}{5}$$

13. Write in lowest terms.

$$\frac{126}{210} = \frac{3}{5}$$

14. Write in lowest terms.

$$\frac{-84}{132} = -\frac{7}{11}$$

15. Write in lowest terms.

$$\frac{144}{96} = \frac{3}{2}$$

16. Write in lowest terms.

$$\frac{-120}{-168} = \frac{5}{7}$$

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Core: equivalent fractions

17. Find the missing numerator.

$$? = 18$$

Multiply numerator and denominator by 6.

18. Find the missing numerator.

$$? = 25$$

Multiply numerator and denominator by 5.

19. Find the missing numerator.

$$? = -8$$

Multiply numerator and denominator by 4.

20. Find the missing numerator.

$$? = 21$$

Multiply numerator and denominator by 3.

21. Find the missing numerator.

$$? = 16$$

Multiply numerator and denominator by 4.

22. Find the missing numerator.

$$? = 33$$

Multiply numerator and denominator by 3.

23. Find the missing numerator.

$$? = -25$$

Multiply numerator and denominator by 5.

24. Find the missing numerator.

$$? = 0$$

Multiply numerator and denominator by 9.

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Extension: compare and explain

25. Are these fractions equivalent? Justify.

$$\frac{3}{4} \text{ and } \frac{3}{4}$$

Yes; the reduced values match.

26. Are these fractions equivalent? Justify.

$$\frac{4}{9} \text{ and } \frac{8}{15}$$

No; the reduced values differ.

27. Are these fractions equivalent? Justify.

$$-\frac{3}{5} \text{ and } -\frac{3}{5}$$

Yes; the reduced values match.

28. Are these fractions equivalent? Justify.

$$\frac{2}{3} \text{ and } \frac{3}{4}$$

No; the reduced values differ.

29. Write in lowest terms.

$$\frac{252}{420} = \frac{3}{5}$$

30. Write in lowest terms.

$$\frac{-165}{225} = -\frac{11}{15}$$

31. Explain why dividing only the numerator by 3 changes the value.

$$\frac{9}{12} \neq \frac{3}{12}$$

The original is $\frac{3}{4}$; the new fraction is $\frac{1}{4}$. Divide both parts by the same nonzero factor to preserve value.

32. Can multiplying both numerator and denominator by zero produce an equivalent fraction? Explain.

No. It produces $0/0$, which is undefined.